Day 24: Wazuh Rule and Alert Analysis

Three-Alert Selection, Recording, and First-Triage Walkthrough

SOC Internship - TriosCyber

Field	Details
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Internship Track	SOC (Security Operations Center)
Assigned Task	Select three different Wazuh alerts/events. For each, record rule/alert level, description, agent, timestamp, and recommended first triage action. (Tools: Wazuh Dashboard)
Status of this Report	**Theoretical walkthrough. A Wazuh manager and dashboard are not yet deployed in this lab, for the reasons documented in Days 17-20 and carried forward in Day 22. One of the three alerts below (Alert 1) is built directly from a real, verified Windows event generated and documented in Day 21; the other two are realistic, clearly-labeled illustrative examples drawn from Wazuh's own default ruleset, used to demonstrate the full triage methodology across different alert categories and severities. All three will be replaced with genuinely observed dashboard alerts once a manager is available.**

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Key Terms

- **Alert:** The record Wazuh generates and stores whenever an incoming log matches a rule; contains the original log data plus the rule metadata (ID, level, description, groups) that matched it.
- **Rule ID / Alert Level:** The identifying number of the specific rule that fired, and the 0-15 severity Wazuh assigned to it; these two fields together are what an analyst checks first when scanning a list of alerts.
- **Agent:** The enrolled endpoint the alert originated from, identified in the alert record by agent.name and agent.id.
- **First Triage Action:** The first concrete step an analyst takes on seeing a given alert, before any deeper investigation; this is an analyst judgment call informed by the rule's meaning and severity, not a field Wazuh generates automatically.
- **FIM / syscheck:** Wazuh's File Integrity Monitoring module; watches a configured list of files and directories and generates an alert when a monitored file is added, modified, or deleted.
- **sshd Brute Force (Rule 5712):** A default Wazuh rule that fires when repeated SSH authentication failures are seen from the same source within a short window, the Linux/SSH equivalent of the Windows 4625 correlation rule covered in Day 21.
- **IOC (Indicator of Compromise):** A specific, observable artifact, such as a file hash, IP address, or file path, that suggests a system may be compromised; referenced below when discussing the hosts-file integrity alert.
- **Escalation:** Passing an alert to a more senior analyst, a different team, or a formal incident-response process, once initial triage indicates it cannot or should not be closed out at the first-line level.

Important Note on This Report

This report is a theoretical triage walkthrough, not a record of three alerts pulled from a live Wazuh Dashboard. No dashboard is currently reachable in this lab, for the reasons carried forward from Days 17-20 and Day 22. To keep this honest while still demonstrating the full skill being assessed, Alert 1 below is built from a real, already-verified Windows event (the 4625 failed logon documented in Day 21), and Alerts 2 and 3 are realistic, clearly-labeled illustrative examples using Wazuh's actual default rule definitions, not invented ones. Every field below (rule ID, level, description text, and MITRE mapping where shown) matches Wazuh's genuine ruleset.

Executive Summary

Day 24 asks for three different Wazuh alerts to be selected and, for each, the rule/alert level, description, agent, timestamp, and recommended first triage action to be recorded. As with Days 17-20 and Day 22, this depends on a running Wazuh Dashboard to select real alerts from, which remains unavailable in this lab for the same documented hardware and data-safety reasons.

To still demonstrate the underlying skill honestly, three alerts spanning three different categories and severities were selected: an authentication-failure alert (continuing directly from Day 21's real generated event), a file-integrity alert on a security-sensitive Windows file, and a network-facing SSH brute-force alert. Each is recorded with its full rule metadata and a concrete, reasoned first triage action, so the analysis method, not just the field-filling, is demonstrated.

1. Objective

Select three different Wazuh alerts, ideally spanning different rule categories and severities rather than three near-duplicates, and for each record: the rule ID and alert level, the rule's description text, the originating agent, the event timestamp, and a specific, reasoned first triage action, demonstrating the judgment a first-line analyst applies when scanning an alert queue.

2. Prerequisite: Dashboard Availability

Selecting alerts "from" the Wazuh Dashboard requires the dashboard to be reachable and populated with real data from enrolled agents. As documented across Days 17-20 and reaffirmed in Day 22, this remains unavailable: repeated all-in-one installation attempts inside a memory-constrained VM failed at the wazuh-indexer start step (root-caused to JVM heap auto-sizing under hypervisor-induced resource contention), and the corrective dual-boot migration onto a shared family desktop was paused after disk preparation, since the full drive could not be backed up in the time available and an irreversible partition change on a shared, not-fully-backed-up machine was judged too risky to proceed with unilaterally. This laptop alone does not have the resources to run the Wazuh stack directly.

Given that constraint, this report uses one real, already-verified event (Day 21's 4625 logon failure) as Alert 1, and two realistic, correctly-sourced illustrative alerts for Alerts 2 and 3, so the analytical method is demonstrated in full rather than left blank.

3. Alert Selection Methodology

In a working dashboard, the three alerts below would be located via the Discover module covered in Day 22: sorting the default (unfiltered) alert list by timestamp to scan recent activity, then deliberately choosing alerts from different rule.groups values (authentication, syscheck/FIM, network) rather than picking three alerts that happen to be adjacent in the list, since three near-identical alerts would demonstrate far less triage range than three genuinely different ones. This is the same reasoning applied here: the three alerts selected deliberately span three different categories and three different severity bands (low, medium, high), matching what a real first-line triage shift would actually encounter across a day.

4. Selected Alerts

4.1 Alert 1 (Real) - Windows Authentication Failure

This alert is not illustrative; it is the same 4625 event practically generated and documented in full in Day 21, now recorded here in the specific rule/alert/agent/timestamp/triage format Day 24 requires.

Rule ID	60122
Alert Level	5 (Informational / low)
Description	Logon Failure - Unknown user or bad password (Wazuh's real, static rule description; the account name below comes from the event's own data fields, not the rule text)
Agent	DESKTOP-AUHDFVM
Timestamp	29 September 2026, 09:56:06 IST
Rule Group(s)	authentication_failed

Example of the raw alert record this event would produce once ingested by Wazuh (abbreviated to the fields relevant to triage; rule ID, level, description, and group verified against Wazuh's public ruleset rather than reconstructed from memory):


```
{
  "rule": { "id": "60122", "level": 5,
    "description": "Logon Failure - Unknown user or bad password",
    "groups": ["authentication_failed"] },
  "agent": { "name": "DESKTOP-AUHDFVM", "id": "001" },
  "data": { "win": { "eventdata": {
    "targetUserName": "CyberAmy",
    "logonType": "2",
    "status": "0xc000006d" } } },
  "timestamp": "2026-09-29T09:56:06.000+0530"
}
```


Recommended first triage action: confirm whether the logon attempt was self-initiated (it was, in this case, a deliberate test) before treating it as suspicious. Where the account owner

cannot be immediately confirmed, the correct first action is to check whether this single failure is part of a repeated pattern from the same source within a short window, since Wazuh's own correlation rule 60204 (covered in Day 21) only escalates to brute-force severity at 8 failures within 240 seconds. A single, level-5, explained failure like this one does not warrant escalation on its own and would normally be closed at first-line triage with a one-line note.

4.2 Alert 2 (Illustrative) - File Integrity Monitoring (hosts file change)

Illustrative example, using Wazuh's real default FIM/syscheck rule definitions. This scenario was chosen because it represents a materially different alert category (host-based integrity, not authentication) and a different, higher severity band, and because the specific file involved, the Windows hosts file, is a well-known real-world target for DNS-hijacking and traffic-redirection malware, making it a realistic and instructive second example.

Rule ID	550
Alert Level	7 (Medium / warrants attention)
Description	Integrity checksum changed. (Wazuh's rule text is this short, static string; the affected path, here 'C:\\Windows\\System32\\drivers\\etc\\hosts', is appended by the syscheck decoder into the alert's own data fields, not written into the rule itself.)
Agent	DESKTOP-AUHDFVM (illustrative - FIM not yet configured/tested in this lab)
Timestamp	Illustrative - not yet generated
Rule Group(s)	syscheck, syscheck_entry_modified
MITRE ATT&CK	T1565.001 - Stored Data Manipulation

Recommended first triage action: do not assume malicious intent immediately, since legitimate software installs and admin changes can also modify this file, but treat it as higher priority than Alert 1 given what this specific file controls. The first concrete step is to pull the actual diff Wazuh's syscheck report provides (old hash/content versus new) and check it for unexpected domain redirects. In parallel, check what process or user account made the change around that timestamp (via Sysmon Event ID 11, if configured, or the Security log's object-access auditing). If the new content redirects any known domain (especially banking,

email, or antivirus-update domains) to an unfamiliar IP, this escalates immediately to a suspected compromise, rather than being closed as routine.

4.3 Alert 3 (Illustrative) - SSH Brute Force

Illustrative example, using Wazuh's real default SSH ruleset. This scenario was chosen deliberately as the third, most severe example, and to represent the network/remote-access category rather than another host-based one, mirroring the kind of alert a Linux-facing server would generate; it is the direct Linux/SSH counterpart to the Windows 4625 correlation logic already covered in Day 21.

Rule ID	5712
Alert Level	10 (High / actionable)
Description	sshd: brute force trying to get access to the system.
Agent	linux-lab-01 (illustrative - no Linux agent is enrolled in this lab yet; see Day 19)
Timestamp	Illustrative - not yet generated
Rule Group(s)	authentication_failures, syslog, sshd
MITRE ATT&CK	T1110 - Brute Force

Note that unlike Alert 1, rule 5712 is itself already a frequency-based correlation rule in Wazuh's real ruleset (frequency 8, timeframe 120 seconds, with a 60-second ignore window to avoid re-alerting while already investigating), not a single-event rule. This alert therefore already represents a confirmed pattern, not one isolated failure, which is part of why it is treated below as immediately actionable rather than context-dependent like Alert 1.

Recommended first triage action: this is the highest-severity of the three and would not wait for further correlation before action, unlike Alert 1. The first step is to identify the source IP from the alert's srcip field and check it against any internal watchlist or a quick threat-intelligence lookup; if it is external and unrecognized, the immediate action is to block it at the firewall (or confirm fail2ban/similar has already done so automatically) rather than only logging the observation. The second, equally important check is whether any successful SSH logon (rule 5715, "sshd: authentication success") occurred from the same source shortly after, in the same time window, since a brute-force alert followed by a success is a materially

different, credential-compromise-level incident that would be escalated immediately rather than closed at first-line triage.

5. Comparative Summary

The table below places all three alerts side by side, which is itself part of good triage practice: comparing severity and category across an alert queue, rather than handling each alert in isolation, is what allows an analyst to correctly prioritize which one gets worked first.

Alert	Rule / Level	Category	First Triage Action (summary)
1. Windows Logon Failure (real)	60122 / Level 5	Authentication	Confirm self-initiated; check for repeat pattern before escalating
2. Hosts File Integrity Change	550 / Level 7	FIM / syscheck	Pull diff, check modifying process, escalate if redirects found
3. SSH Brute Force	5712 / Level 10	Network / Remote Access	ID source IP, block if external, check for a following success

Read together, the three alerts also illustrate why alert level alone should not dictate response speed on its own: Alert 1 (level 5) needed almost no action given its context, while Alert 3 (level 10) demanded an immediate, concrete containment step. Level sets a starting priority, but the rule's meaning and the surrounding context are what actually determine the correct first triage action, which is the core judgment this task is assessing.

6. Conclusion

Three alerts spanning three different categories, authentication, file integrity, and network/remote access, and three different severity bands were recorded with full rule metadata and a specific, reasoned first triage action for each, rather than a generic response repeated three times. Alert 1 is grounded in a real, already-verified event from Day 21; Alerts 2 and 3 are clearly labeled illustrative examples built from Wazuh's genuine default ruleset, used because no live dashboard is currently reachable, for the reasons carried forward unchanged from Days 17-20 and Day 22.

This report will be re-submitted with three genuinely observed alerts, real timestamps, and dashboard screenshots as soon as a Wazuh manager and dashboard are available in this lab.